

Project Proposal

Project Evolution

Please list out changes in the directions of your project if the final project is different from your original proposal (based on your stage 1 proposal submission).

Answer: The project maintained its core functionality, but we encountered significant challenges with our recommendation system that required resolution. As a result, we decided to replace part of the recommendation algorithm with vector similarity search implemented in Python. We successfully integrated all note combinations and incorporated a new trigger mechanism.

Project Achievement

Discuss what you think your application achieved or failed to achieve regarding its usefulness.

Answer: Our application successfully helped users optimize search functionality for articles. This feature proved particularly valuable, and we were pleased with the results. We enhanced the user experience by implementing additional query features to create a coherent process from trigger mechanisms to advanced queries and supplementary features.

Data Schema

Discuss if you change the schema or source of the data for your application.

Answer: We maintained the original schema structure but enhanced it by adding a new ArticleStats table. We implemented a trigger that updates this ArticleStats schema, allowing us to track and query important metrics from the backend.

Design Changes

Discuss what you change to your ER diagram and/or your table implementations. What are some differences between the original design and the final design? Why? What do you think is a more suitable design?

Answer: Our table implementations proved more effective than initially anticipated. We found that ER diagrams serve primarily as conceptual tools, whereas practical database development often requires writing SQL queries first to properly understand table relationships and optimize implementation.

Functionality Modifications

Discuss what functionalities you added or removed. Why?

Answer: We faced time constraints in developing our recommendation search feature, which led us to implement vectorized search functionality instead. Additionally, we incorporated trigger

mechanisms into our table structure to automatically maintain updated data for the ArticleStats table.

Advanced Database Integration

Explain how you think your advanced database programs complement your application.

Answer: The project presented several technical challenges, including connecting the backend to the frontend, developing methods to store data effectively, and implementing triggers in a production-oriented manner rather than an academic context.

Technical Challenges

Each team member should describe one technical challenge that the team encountered.

Answer: Building a complete application from start to finish presented significant challenges, but we ultimately succeeded. Anu worked on refining and filtering frontend features to facilitate integration. Arnav collaborated with Anu and focused on developing frontend endpoints for data storage to enable seamless backend-to-frontend integration. Mike managed most of stage three, creating and integrating tables to optimize query performance. Shubham developed the Keyword Search functionality, implementing a machine learning algorithm to enhance search efficiency, and designed an advanced trigger system with complex constraints and advanced queries.

Project Documentation

This should be sufficiently detailed such that another future team could use this as helpful advice if they were to start a similar project or where to maintain your project.

Answer: Our repository contains various components organized logically. We manage dependencies through the venv package manager, which future teams should familiarize themselves with. The core module serves as the main orchestrator, handling primary logic and storing data models referenced by endpoints and vector search functionality. Our recommender folder contains the SearchBar and advanced triggers/queries used in endpoints. The SQL folder houses all our queries and advanced queries in a modular "plug and play" format, alongside the Flask APIs that power our backend.

Additional Changes

Are there other things that changed comparing the final application with the original proposal?

Answer: No additional changes beyond those mentioned above.

Future Work

Describe future work that you think, other than the interface, that the application can improve on.

Answer: Future development should focus on implementing a robust recommendation system and scaling the application to support multiple users. Currently, the application functions as a single-user system, but expanding to accommodate multiple users would enhance both search functionality and recommendation capabilities through increased interaction data.

Team Collaboration

Describe the final division of labor and how well you managed teamwork.

Answer: Shubham and Michael Yan collaborated on backend features, database ER diagrams, and schema development. Shubham focused on all backend features integrated into the app. Michael focused on query development, setup, and data insertion processes. Arnav and Anu handled all frontend components. While coordinating efforts was challenging due to competing academic deadlines and exams, we successfully delivered a functional system. Though some frontend to backend connections remain incomplete, the majority of the system functions effectively.